
HOMEWORK DAY 27 – *Representing functions by power series §11.9*

1. §11.9: 1

2. §11.9: 2

3. Find power series representations for the following functions about $x = a$ by manipulating known series. State their radius and interval of convergence.

(a) $f(x) = \frac{1}{1+x}, a = 0$

(b) $f(x) = \frac{x}{1+x^2}$, $a = 0$

(c) $f(x) = \frac{1}{1+x}$, $a = 3$

(d) $f(x) = \tan^{-1}(x)$, $a = 0$

(e) $f(x) = \tan^{-1}(x^4)$, $a = 0$

(f) $f(x) = \frac{2}{3-x}, a = 0$

4. (a) Find a power series representation of $f(x) = \frac{1}{(1-x)^2}$ about $a = 0$ (hint: use differentiation of the geometric series).

- (b) Use your result above to find a power series representation of $f(x) = \frac{x}{(2-x)^2}$ about $a = 0$.

5. *Application: approximating functions.*

(a) Use the geometric series to find a power series representation for $\ln(1 + x)$ about $x = 0$.

(b) Find a power series representation for $\ln(1 + x^2)$ about $x = 0$.

(c) Use the resulting series to estimate the value of $\ln(1.01)$ to within 10^{-7} (roughly single machine precision).